Scotsman®

Instructions

Installation and Operation of the Prodigy Smart-Board™, KSB

Scotsman's Advanced Feature Smart-Board is an optional add on electronic device that can be applied to most Prodigy models. It can be used:

- With the standard controller
- With the standard controller and the SmartLock Out Control (KSL)
- With the standard controller and the Vari-Smart™ Ice Level Control (KVS)
- With the standard controller, and both the KSL and the KVS

Smart-Board abilities include:

- USB connection to Scotsman's Prodigy TechTool software
- Data Logging
- Data Display

Kit Contents (permanent mount)

- Smart-Board (AFB), mounting screws, overlay label, connecting cable

Installation:

1. Depress and hold the Off button until the machine shuts Off (Status light will go out).

2. Disconnect electrical power from the ice machine.

3. Remove front panel.

4. Locate blank lower part of overlay panel (label). Score perforation with knife, lift up bottom corner of label and peel the blank portion away from the control box cover up to the perforation. Flex several times and remove the label from the control box cover.

5. Attach supplied overlay to control box door in place of the one removed in step 4.

6. Remove screw holding control cover to control box, swing control & cover open.

7. Install AFB board to bottom of control box cover. Secure to cover with the supplied screws.

8. Connect supplied wire from AFB J1 to main controller Accessory.

9. Close the control box cover.

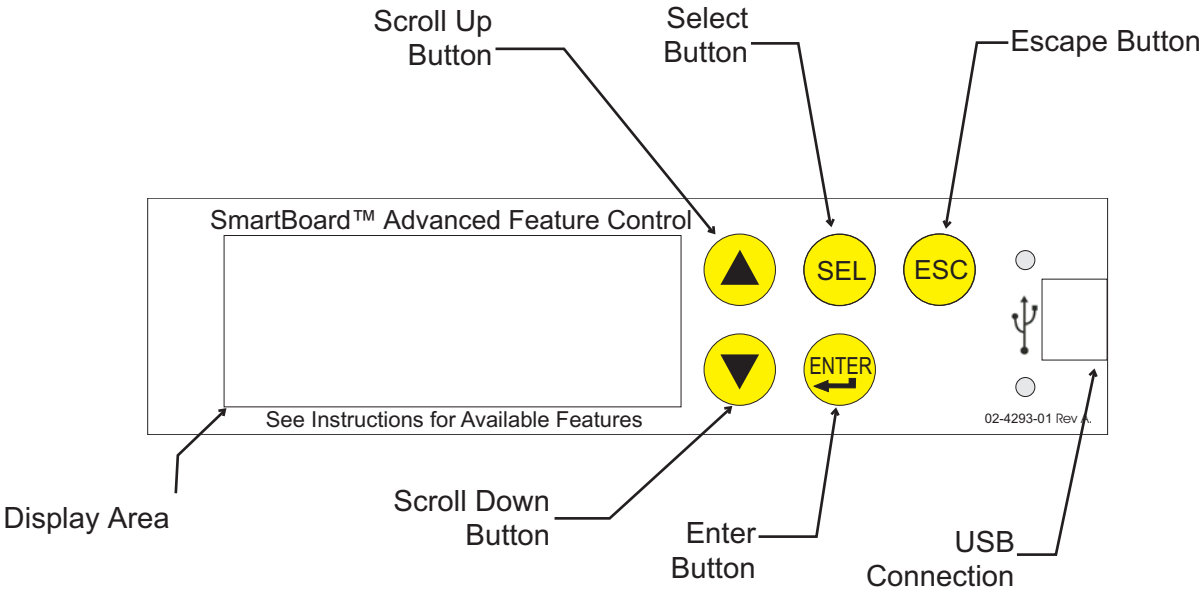
10. Reconnect electrical power. Display will show time and date (US Central Time). See time set section for instructions on changing the time.

17-3137-01

Use of Smart-Board Buttons:

Scroll Up: Changes the display to a menu item higher on the menu list or goes up one number on a setting

Scroll Down: Changes the display to a menu item lower on the menu list or goes down one number on a setting



Select Button: Use to make changes to settings.

Enter Button: Changes display to a sub menu list.

Escape Button: Changes display to the main menu.

The Smart-Board can display **Warnings** and **Data**.

Data Available:

- Time, Date
- Average freeze time
- Minimum freeze time
- Maximum freeze time
- Average harvest time
- Minimum harvest time
- Maximum harvest time
- Diagnostic code with timestamp
- Compressor run time
- Freeze cycles
- Flush level used
- Water quality
- Operational mode
- Water temperature
- Discharge temperature
- Voltage from the transformer
- Bin stat input status

Warnings - will appear in display after malfunction

- Self Test Fail
- Long Freeze Pend
- Long Freeze Err
- Long Harvest Err
- Check Water
- High Temp Error
- Sump Temp Sensor
- Disch Temp Sensor
- Min Freeze Pend
- Min Freeze Error
- Check Water Warn
- Long Freeze Warn
- Long Harv Warn
- High Temp Warn

Communication Features:

USB: There is a USB connection on the front of the SmartBoard. It can be used by a laptop or other PC type computer to read, download or log data. Scotsman software is included on the CD-ROM. Instructions for use of that software are also on the CD-ROM in Adobe Acrobat PDF format. Updates for the CD-ROM software may be available. Check at www.scotsman-ice.com for more information.

Other Features:

7 Day Programmable Ice Level Control is available when the optional Vari-Smart adjustable ice level control is installed on the Prodigy controller. Instructions for programing are included in these instructions.

Note: Some features not available if installed on a cuber that has Rev 1 software. Rev 2 use began approximately March 2007.

If a Smart-Board and a Vari-Smart are used with a Rev 1 software controller:

- Bin level control cannot be disabled by setting programmable value to 0. Setting to 0 uses the current switch setting, whatever it happens to be at.*
- Bin Full LED is not controlled by the Smart-Board, but by the disabled selector switch. This means it can be on or off with no correlation to actual level.
- In the Smart-Board menu, there's a value called Bin Level Setpoint, it'll show the disabled selector switch value instead of the programmable value.

Software Rev can be identified using the Smart-Board or there is a white sticker on the cuber controller, next to the Bin Stat connector. That sticker lists the software rev (as in S.W. Rev: 2).

*** Recommendation when using Rev 1: Set the Vari-Smart control knob to Full (arrows point at each other).**

Menu Tree

Date - preset
Time - preset to Central Time
Warnings
- No warnings
- See prior page for warning list

Base Faults
- Fault code 1:
- Fault code 2:
- Fault code 3:
- Fault code 4 :
- Fault code 5 :
- Fault code 6 :
- Fault code 7 :
- Fault code 8 :
- Fault code 9 :
- Fault code 10 :

Adv (advanced) Faults, description of fault and time and date of occurrence
- Self test failure
- Long Freeze Pend
- Long Freeze Strikeout
- Long Harv Pend
- Long Harvest Strikeout
- Check Water
- High Temp Error
- Disch Temp Error
- Sump Temp Sensor
- Discharge Temp Sensor
- Min Freeze Pend
- Minimum Freeze Strikeout

Status
- Discharge Temp
- Sump Temp
- Board Voltage
- Bin Level
- Bin Setpoint
- Freeze Timer
- Harvest Timer
- Freeze Counter
- Water Quality
- Flush Used
- Long Frz Strike
- Long Hrv Strike
- Min Frz Strike
- Pwr Interrupts
- Bin Stat
- Disch Frz Set

Cleaning
- Clean interval
- Next Clean Due
- Last Clean
- Flush level
  - Set Flush Level

Performance
- Percent run time:
- Min Freeze Time
- Max Freeze Time
- Avg Freeze Time
- Min Harvest Time
- Max Harvest Time
- Avg Harvest Time
- Clear History

Test
- Water Test
  - Water fill time
  - Esc to cancel test

Timers
- Compressor run time
- Comp resettable
  - Press enter to reset
- Pwr up time
- Pwr resettable
  - Press enter to reset

Revision
- AFB SW Revision
- Controller SW
- US Bin Level SW
- AFB Hardware Rev
- Controller HW

Setup
- Date
  - Set date
- Time
  - Set time
- Model Number
  - Set model number
- Serial Number
  - Set serial number
- Manufacturer
- Equipment Name
- Manufacture date
  - Set Manufacture Date
- Install Date
  - Set Install Date
- Contact Name
  - Set Contact Name
- Contact Phone Number
  - Set contact phone
- Audible alert
  - Set audible alert on / off
- Clear current log file
  - Press Select to clear log
- Clear fault history
  - Press Select to clear fault code
- Logging rate
  - Set logging rate
- Fill time warning xxx seconds
  - Set fill time warning
- Freeze time warning xx minutes and seconds
  - Set freeze time warning
- Harvest time warning minutes and seconds
  - Set harvest time warning
- Discharge temp warning in degrees F
  - Set discharge temp warning

PGM Bin Level*
- Bin Level Ctrl
  - Set Bin Level Ctrl On Off
- Monday time 1
- Monday level 1
- Monday time 2
- Monday level 2
- Monday time 3
- Monday level 3
- Monday time 4
- Monday level 4
- Tuesday time 1
- Tuesday level 1
- Tuesday time 2
- Tuesday level 2
- Tuesday time 3
- Tuesday level 3
- Tuesday time 4
- Tuesday level 4
- Wednesday time 1
- Wednesday level 1
- Wednesday time 2
- Wednesday level 2
- Wednesday time 3
- Wednesday level 3
- Wednesday time 4
- Wednesday level 4
- Thursday time 1
- Thursday level 1
- Thursday time 2
- Thursday level 2
- Thursday time 3
- Thursday level 3
- Thursday time 4
- Thursday level 4
- Friday time 1
- Friday level 1
- Friday time 2
- Friday level 2
- Friday time 3
- Friday level 3
- Friday time 4
- Friday level 4
- Saturday time 1
- Saturday level 1
- Saturday time 2
- Saturday level 2
- Saturday time 3
- Saturday level 3
- Saturday time 4
- Saturday level 4
- Sunday time 1
- Sunday level 1
- Sunday time 2
- Sunday level 2
- Sunday time 3
- Sunday level 3
- Sunday time 4
- Sunday level 4

Network Configuration***
- IP Address
- Subnet mask
- Default gateway
- DHCP Enable
- Update IP Address**
- Update Subnet mask**
- Update default Gateway**
- Update DNS**
- Update DHCP**
- Update network on next power cycle**

* all under PGM Bin Level also have a Set menu
** also have a Set menu
*** Does not apply to this version Smart-Board.

Advanced Fault Definitions

Self test failure

The controller checks for proper operation at power up. If the check shows a problem, this warning or fault will be displayed.

Long Freeze Pend

If the ice machine fails to make ice within the maximum time limit, the controller will note that and display this warning or fault while it is attempting another freeze cycle.

Long Freeze Strikeout

If the ice machine fails to make ice within the maximum time limit for a third consecutive time, this warning or fault will be displayed and the machine will be shut down.

Long Harv Pend

If the ice machine fails to release ice within the maximum time limit, the controller will note that and display this warning or fault while it is attempting another freeze cycle

Long Harvest Strikeout

If the ice machine fails to release ice within the maximum time limit for a third consecutive time, this warning or fault will be displayed and the machine will be shut down.

Check Water

If the water level sensor does not sense a full reservoir during the maximum time limit, this warning or fault will be displayed. The machine will automatically attempt to fill with water.

High Temp Error

If the discharge temperature exceeds 250 degrees at any time, the controller will shut the machine down and display this warning or fault.

Sump Temp Sensor

The water temperature sensor's resistance varies with the water temperature. If the resistance is beyond what the sensor's capability is, this warning or fault is displayed.

Discharge Temp Sensor

The discharge temperature sensor's resistance varies with the refrigerant temperature. If the resistance is beyond what the sensor's capability is, this warning or fault is displayed.

Min Freeze Pend

If the controller senses finished ice thickness before the minimum freeze time has elapsed, this warning or fault will be displayed.

Minimum Freeze Strikeout

If the controller senses finished ice thickness before the minimum freeze time has elapsed three cycles in a row, this warning or fault will be displayed if the machine and the machine will be shut down.

Smart-Board Button Use:

Menu Groups: Push and release the down arrow key to scroll down to the next group.

```
Date:  12-20-2006
Time:  03:33:10PM
```

Date - preset

Time - preset to Central Time

Warnings

Base Faults

Adv (advanced) Faults

Status

Cleaning

Performance

Test

Timers

Revision

Setup

PGM Bin Level

Network Configuration (N/A this version)

```
Warnings
Base Faults
```

```
Adv Faults
Status
```

```
Cleaning
Performance
```

```
Test
Timers
```

```
Revision
Setup
```

```
Setup
PGM Bin Level
```

```
PGM Bin Level
Network Config
```

Within each group are several screens of information or settings, like times, that can be changed.

Date and Time Groups: No submenus are available.

Warnings: Press and release the Enter button to see information on current Warnings.

```
Warnings
Base Faults
```

Press and release ESC to return to the prior menu.

```
No Warnings
```

Base Faults: Press and release the Down arrow to underline the B in Base Faults, then the Enter button to see in the display:

Most recent failure (labeled 0) and how long ago it occurred (in hours), then press and release the down arrow to see:

```
Warnings
Base Faults
```

Second to most recent failure (labeled 1) and how long ago it occurred (in hours), then press and release the down arrow to see the third, fourth, fifth, and so on up to ninth where the list ends.

```
Warnings
Base Faults
```

If there are no errors, the screen will display End of Errors.

```
End of Errors
```

Press and release the escape button to return to the main menu tree.

ESC

Press and release the down arrow key to underline the A in Advanced Faults.

```
Base Faults
Adv Faults
```

Advanced Faults: Press and release the Enter button to see in the display:

ENTER

Most recent failure and the exact time it occurred. Pressing and releasing the down arrow cycles through the other failures back to the oldest.

Several examples are listed to the right.

```
Long Harv Pend
04-15-07:12:02AM
```

```
Long Freeze Pend
04-13-07:02:16PM
```

```
Check Water
04-07-07:10:56AM
```

```
High Temp Error
03-13-07:08:59PM
```

```
Min Freeze Pend
02-14-07:06:31AM
```

```
Sump Temp Sensor
01-15-07:11:17PM
```

At the end of the list the display will show directions to go back to the main menu.

```
up arrow = back
esc = main menu
```

Press and release the escape button to return to the main menu tree.

ESC

Press and release the down arrow to underline the S in Status.

Status: Press and release the Enter button to see:

Adv Faults
Status

Discharge Temp in degrees F., Then press and release the Down arrow key to see:

ENTER

Discharge Temp:
157

Sump Temp in degrees F. Then press and release the Down arrow key to see:

Sump Temp:
38

Board Voltage - from the transformer. Then press and release the Down arrow key to see:

Board Voltage:
14

Bin Level number. Use with Vari-Smart control. Displays level currently sensed. Will display 255 when no Vari-Smart present. Then press and release the Down arrow key to see:

Bin Level:
13

Bin set point: Used with Ultrasonic control. Then press and release the Down arrow key to see:

Bin Setpoint:
9

Freeze Timer: Freeze time. Then press and release the Down arrow key to see:

Freeze Timer:
00:00

Harvest Timer:: Harvest time. Then press and release the Down arrow key to see:

Harvest Timer:
00:00

Freeze Counter: Then press and release the Down arrow key to see:

Freeze Counter:
0

Water Quality.: Measurement of the conductivity of the reservoir water. Typically between 20 and 60, lower numbers mean higher mineral content. Then press and release the Down arrow key to see:

Water Quality:
0

Flush Used: The WaterSense system has selected this purge setting. Will read 255 if no water in sump. Then press and release the Down arrow key to see:

Flush Used:
0

Long Freeze Strike: Long Freeze Strike number. Number of long freeze errors in memory. Then press and release the Down arrow key to see:

Long Frz Strike:
0

Long Harvest Strike: Long Harvest Strike number: Number of long harvest errors in memory. Then press and release the Down arrow key to see:

Long Hrv Strike:
0

Min Frz Strike: Minimum freeze strike number Then press and release the Down arrow key to see:

Min Frz Strike:
0

Pwr Interrupts: Number and time of power interruptions. Then press and release the Down arrow key to see:

Pwr Interrupts:
0

Bin Stat: Open or Closed. Open is normal when no bin thermostat is attached or there is no ice on a thermostat. Then press and release the Down arrow key to see:

Bin Stat:
Open

Disch Frz Set: In degrees F. Shows the discharge temperature recorded as a set up number. The set up number is used for determining how long the fan is off at the end of the freeze cycle.

Disch Frz Set:
0

When done with Status, press and release the ESC button.

ESC

Push and release the Down arrow to put the line under the C in Cleaning. Then push and release the Enter button to see.

Status:
Cleaning

ENTER

Cleaning. Press and release the Enter button to see:

The Clean Interval. Then press and release the Down arrow to see:

Clean Interval:
HR

The Next Clean Due in x HRS. Then press and release the Down arrow to see:

Next Clean Due
in HRS

Last Clean: x HR Ago.

Then press and release the Down arrow to see:

Last Clean:
HR Ago

Flush Level: Set to Auto or 1, 2, 3, 4 or 5.

SEL

Push and release the SEL arrow key to enter flush level set mode.

Flush Level:
1

Push and release the Up or Down arrow keys to change flush level.

Set Flush Level:
3

Push and release the Enter key to set the new flush level.

ENTER

ESC

Then press and release the ESC button.

Push and release the Down arrow to put the line under the P in Performance.

Cleaning
Performance

Then push and release the Enter button to see:

Performance

Percent run time. Then press and release the Down arrow to see:

```
Percent    run
time
```

Min Freeze Time. Then press and release the Down arrow to see:

```
Min Freeze Time
00:00
```

Max Freeze Time. Then press and release the Down arrow to see:

```
Max Freeze Time
00:00
```

AVG Freeze Time. Then press and release the Down arrow to see:

```
AVG Freeze Time
00:00
```

Min Harvest Time. Then press and release the Down arrow to see:

```
Min Harvest Time
00:00
```

Max Harvest Time. Then press and release the Down arrow to see:

```
Max Harvest Time
00:00
```

AVG Harvest Time.

```
AVG Harvest Time
00:00
```

Press and release the Down arrow open last time to enter the Clear History screen. Press and release the SEL button to clear the performance history.

```
Clear History
```

When done with Performance, press and release the ESC button.

Push and release the Down arrow to put the line under the T in Test. Then press and release the Enter button to see:

```
Performance
Test
```

Test. Press and release the SEL button to begin a water test. The time to fill the reservoir will be displayed.

```
Press   select
to   start   water
```

When done with Test, or to cancel it, press and release the ESC button.

Push and release the Down arrow to put the line under the T in Timers. Then press and release the Enter button to see:

```
Test
Timers
```

Timers. Push and release the Enter button to see

ENTER

Compressor run time. Then press and release the Down arrow to see:

```
Compressor Run:
HR
```

Compressor run resettable. Press the Down arrow to go to the next line or Press SEL to enter reset mode.

SEL

```
Comp Resettable:
0HR
```

Press Enter to reset compressor run time to 0

ENTER

```
Press enter to
clear counter
```

Press the Down arrow to go to Power up time. Then press and release the Down arrow to see:

```
Pwr Up Time:
HR
```

Power on resettable. Press the Down arrow to go to the next line or Press SEL to enter reset mode.

SEL

```
Pwr Resettable:
HR
```

Press Enter to reset Power on time to 0

ENTER

When done with Timers, press and release the ESC button.

ESC

Push and release the Down arrow to put the line under the R in Revision. Then push and release the Enter button to see:

```
Timers
Revision
```

Revision. AFB SW Rev number.

ENTER

Then press and release the Down arrow to see:

Controller SW (software rev number) Then press and release the Down arrow to see:

```
AFB SW Rev
```

US Bin Level (Vari-Smart) software revision. Then press and release the Down arrow to see:

```
Controller SW
```

```
US Bin Level SW
140
```

AFB Hardware Rev (Smart-Board revision)

Then press and release the Down arrow to see:

Controller HW (hardware rev number).

When done with Revisions, press and release the ESC button.

Push and release the Down arrow to put the line under the S in Setup.

Then push and release the Enter button to see:

Setup:

View the Date or change it.

To Set Day, Month and Year

Press SEL key to get to Setup screen

Push and release the SEL key to move to another underlined number.

Push and release the Up or Down arrow key to change the marked character.

Push and release the Select key to move to the next character, repeat prior step to change the character.

When done, push and release the Enter key.

Then press and release the Down arrow to view the time or change it.:

To Set Time

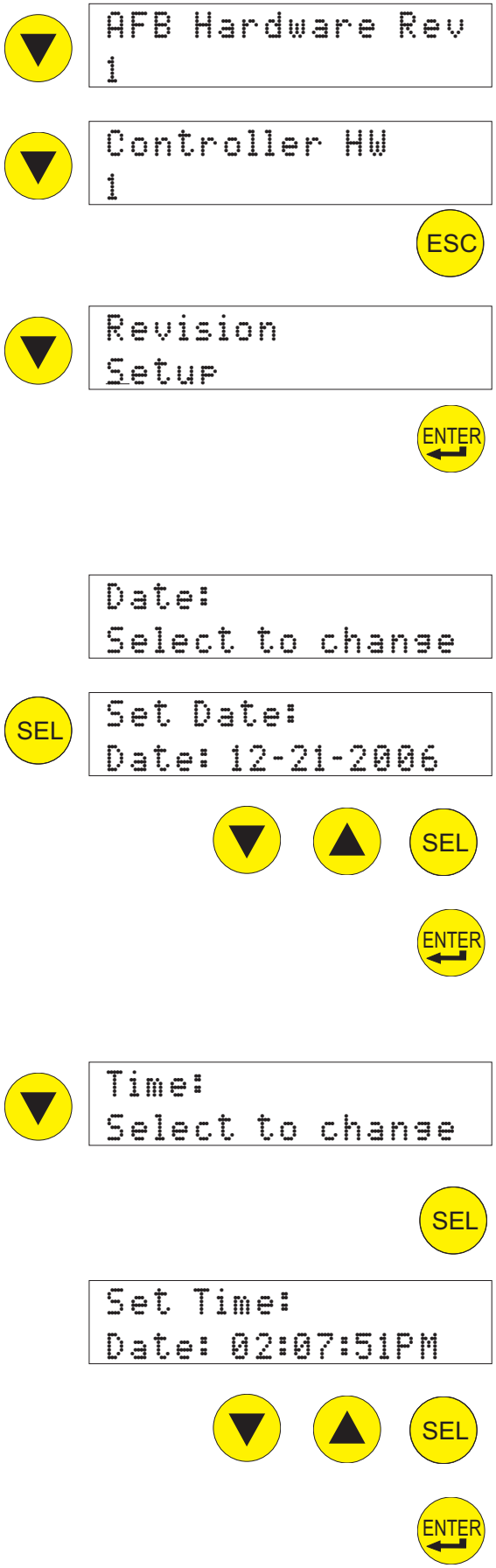
Press SEL key to get to Setup screen

Push and release the SEL key to move the underline to another number.

Push and release the Up or Down arrow key to change the marked character.

Push and release the Select key to move to the next character, repeat prior step to change the character.

When done, push and release the Enter key.



Then press and release the Down arrow to view the

Model number.

Then press and release the Down arrow to view the

Serial number

Then press and release the Down arrow to view the

Manufacturer

Then press and release the Down arrow to view the

Equipment Name

Then press and release the Down arrow to view the

Manufacture date

Then press and release the Down arrow to view the

Install date

Then press and release the Down arrow to view the

Contact name

Then press and release the Down arrow to view the

Contact phone number. Then press and release the Down arrow to view the

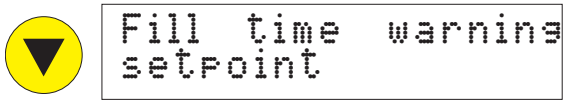
Audible Alert.

Then press and release the Down arrow to view the. Then press and release the Down arrow to view the screen to clear the current log file.

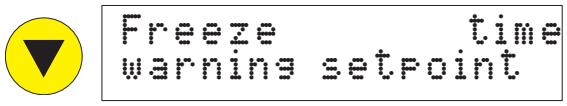
Then press and release the Down arrow to view the Clear Fault History file. Then press and release the Down arrow to view the

Logging rate. Then press and release the Down arrow to view the

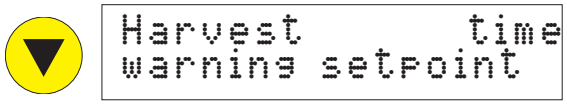
Fill time warning. Then press and release the Down arrow to view the



Freeze time warning. Then press and release the Down arrow to view the



Harvest time warning. Then press and release the Down arrow to view the



Discharge temp warning.

Any of the above can be modified by changing the settings as noted below. The warning set points can be adjusted to match local conditions, so that when they change the Smart-Board provides a notice of the change.

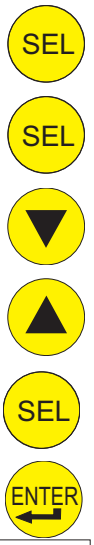
To Change Setup Settings:

Press SEL key to get to Setup screen. Push and release the SEL key to move the underline to another number.

Push and release the Up or Down arrow key to change the marked character.

Push and release the Select key to move to the next character, repeat prior step to change the character.

When done, push and release the Enter key.

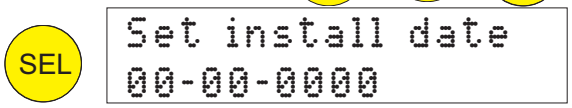


Example 1: Set Install Date

Push the Down arrow key until Setup is visible and the S is underlined. Press Enter.

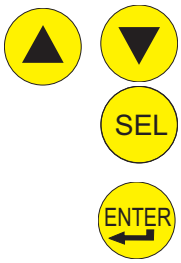
Repeatedly push and release the Down key until the Install Date screen appears. Press SEL key to get to Setup screen

Push and release the SEL key to move the underline to another number. Push and release the Up or Down arrow key to change the marked character.



Push and release the Select key to move to the next character, repeat prior step to change the character.

When done, push and release the Enter key.



To Set the 7 Day Programmable Ice Level - only usable with Vari-Smart control. There are four times and levels available for each day of the week.

Use Up or Down Arrow keys to scroll to **PGM Bin Level** screen. When the P in PGM is underlined, press and release the Enter key.

Bin Level Ctrl is on the screen. Push and release the SEL button to get to the screen that can toggle the Vari-Smart control on or off (up or down arrow keys).

The control must be set to OFF to adjust bin level..

Monday Time 1 will show on the screen.

Press and release the SEL key to begin setting Monday Time 1. **Set Monday Time** appears in the display.

01:30AM might be displayed. To adjust, position the underline mark under the character to be changed. Repeat pushing the SEL button to move the underline. Push and release the up or down arrow key to change the number or letter one time.

Push Enter once to change the screen back to Monday Time 1. Push Down once.

Monday Level 1 is in the display. Push SEL once, Set Monday Level 1 shows in the display. Rotate the Vari-Smart knob one click at a time until the desired level appears. A delay is normal. Note: smaller numbers = higher ice level. Push Enter one time to set the level. "Off" means the switch is set for maximum ice level. **Off not available with SW Rev 1 controller**. Rev level on sticker on back of controller.

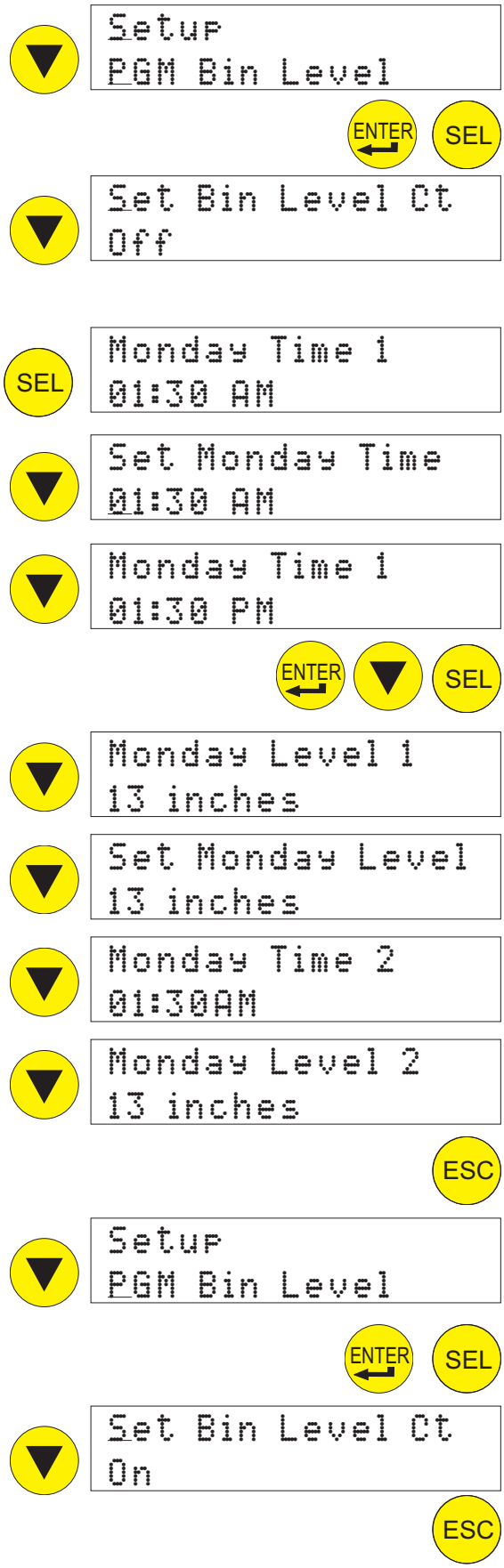
Push the Down arrow key to go to **Monday Time 2**, and set that time. Push ESC once to change back to Monday Time 2, and **Monday Level 2** will show in the display. Again, rotate the Vari-Smart knob one click at a time until the desired level appears.

Repeat for all time and level settings.

When done, push and release the ESC to return to the PGM Bin Level screen. When the P in PGM is underlined, press and release the Enter key.

Bin Level Ctrl will show on the screen. Push and release the SEL button. Push and release an arrow key to change the setting from Off to On.

Push and release ESC when done.



Example: Unit on a tall bin (44" or more). Low ice levels during the week, high ice levels during the weekend. Off = highest ice level.

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Time 1	1:00 AM	1:00 AM	1:00 AM	1:00 AM	1:00 AM	1:00 AM	1:00 AM
Level 1	32	32	32	32	32	Off	Off
Time 2	11:00 AM	11:00 AM	11:00 AM	11:00 AM	11:00 AM	11:00 AM	11:00 AM
Level 2	32	32	32	32	32	Off	14
Time 3	3:00 PM	3:00 PM	3:00 PM	3:00 PM	3:00 PM	3:00 PM	3:00 PM
Level 3	32	32	32	32	32	Off	32
Time 4	7:00 PM	7:00 PM	7:00 PM	7:00 PM	7:00 PM	7:00 PM	7:00 PM
Level 4	32	32	32	32	Off	Off	32

Note: Units on dispensers or short bins should not set their level other than 9 or Off - setting too low will result in no ice.

Record settings here for future reference:

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Time 1							
Level 1							
Time 2							
Level 2							
Time 3							
Level 3							
Time 4							
Level 4							

Diagnostic Example: Look up Average Freeze Time:

Use Up or Down arrow keys to scroll to Performance screen.

Push and release the Enter arrow.

Push and release the Up arrow until the Average Freeze Time screen appears.

When done, push and release the ESC button until the Main screen appears.

Supplied Software

Description:

Scotsman Prodigy Tech Tool 1.0 is a software program designed to access the Prodigy ice machine Smart-Board. It can read and display the data in the controller. The data is converted to chart form and can then be saved and / or printed. It is on the CD-ROM and must be installed onto the PC that will be used to connect through the USB port of the Smart-Board.

Installation and Use

Requirements:

- Windows XP.
- 40 MB disk space minimum. More will be needed if data logging is used.
- Desktop or Laptop PC with a USB port.
- Live ice machine with Smart-Board (to install USB driver)

Software Installation:

Pre-installation: The Smart-Board must be UNPLUGGED from the the PC.

1. Insert the CD into the computer's CD-ROM drive.

2. Follow the program installation instructions.

3. The installation will place 2 icons on the desktop, Prodigy Charting and Scotsman Prodigy.

4. The installation will also set up a Scotsman Prodigy section under Programs (Start > All Programs > Scotsman Prodigy).

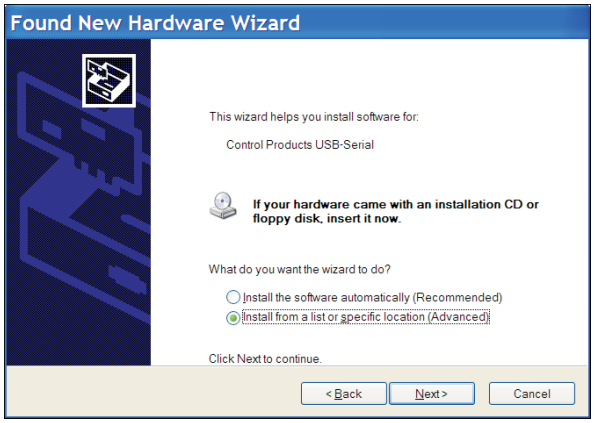
5. After the installation is complete, leave the CD-ROM in the drive.

6. Power up the Smart-Board and plug the USB connector into the PC and the Smart-Board.

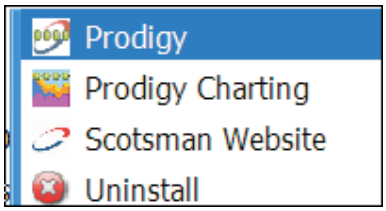
7. The PC will automatically find the Smart-Board and begin the process to install the driver.

8. Select all default settings for installing the device driver. The CD-ROM must still be in the drive.

9. Installation is now complete. Remove the CD-ROM.

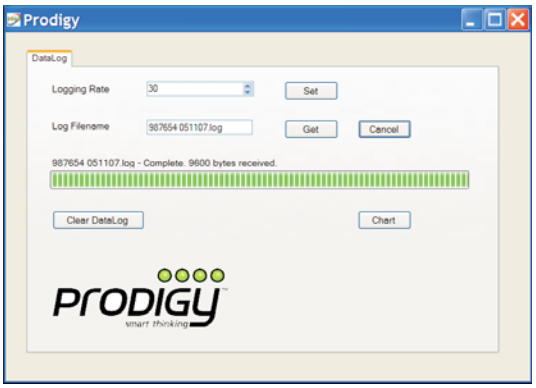


Use:



With the Smart-Board powered and connected to the computer's USB port, Open Scotsman Prodigy:

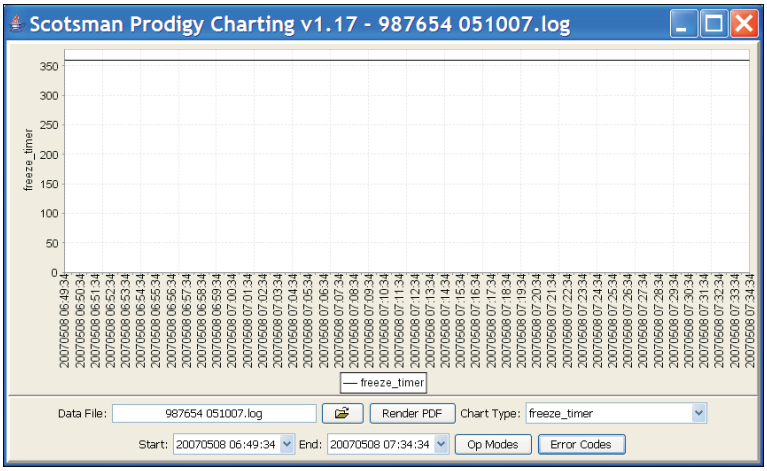
Start, All Programs, Scotsman Prodigy, Prodigy



Click on **Get**. The software will automatically begin to download the information from the Smart-Board. Once that is complete either click on **Chart** or select a new log file to review.

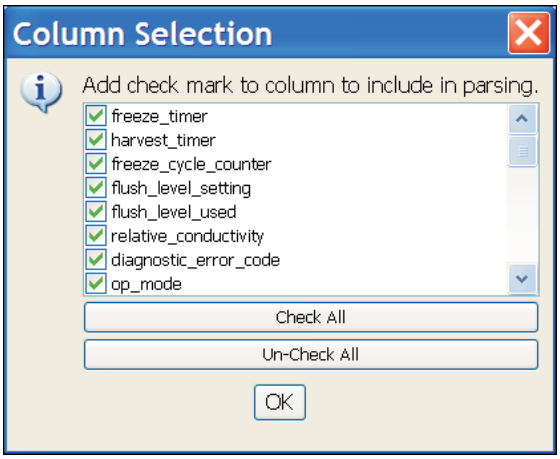
When you click on **Chart**, the Column Selection dialog box will appear. You can select any chart you want to review. The default is all of them. Click on **OK** to go to the next step.

The software will display the Charting information box. You may have to expand it to see the Chart Type selection area on the right.



Use the Chart Type box to display the list of available charts.

Which one of these to use depends upon what the machine situation is. For example, if the machine is down, displaying code 2, indicating a maximum length harvest cycle, it would be good to know the freeze cycle time before the long harvest cycle. If the freeze cycle time is long, it may be that no ice is being made, so none is available to open the curtain during harvest, causing a maximum harvest time code. Checking the Base Faults or Advanced Faults is another way to understand what occurred and when.



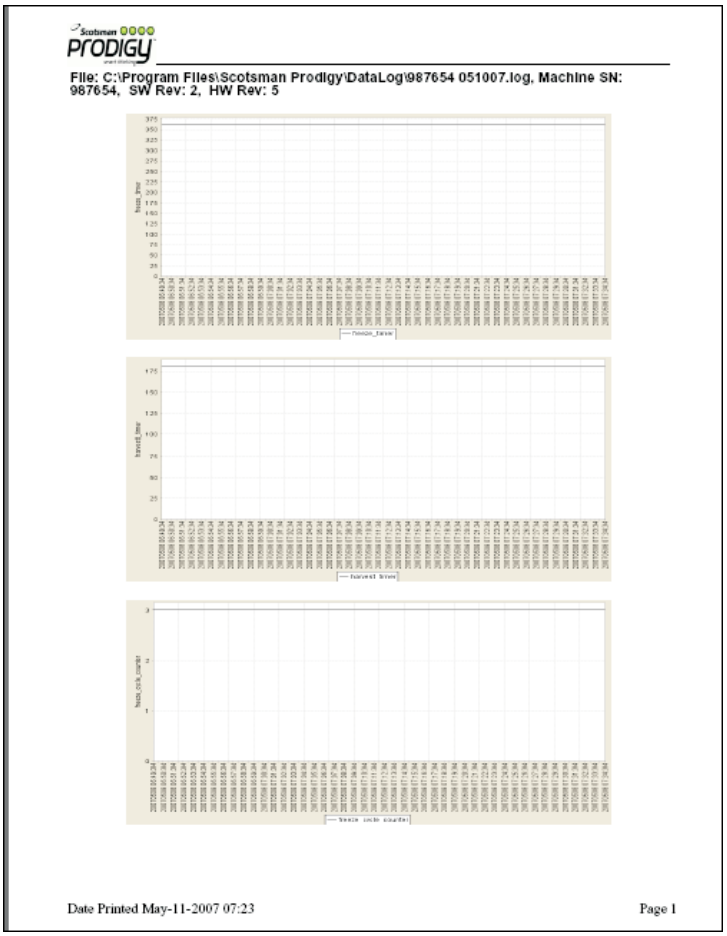
Another example is a complaint of low capacity. The chart on Power up time should show if the machine is on all the time. Then a look at the freeze timer chart will show how often it is cycling. The two will provide a good idea of the machine's ability to produce ice.

At any time clicking on the **Render PDF** button will generate all the charts in PDF format so they can be saved. Once saved they can be printed or emailed.

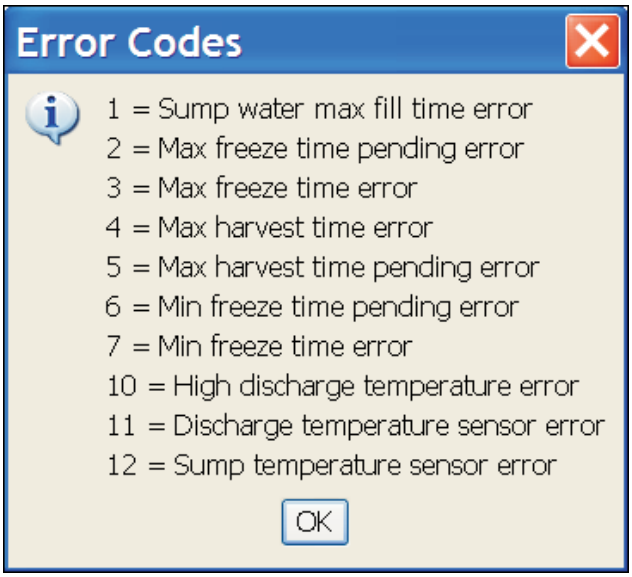
Chart Definitions:

- Freeze timer = Freeze time in seconds.
- Harvest timer = Harvest time in seconds.
- Freeze Counter = Continuous freeze cycles (starts over after bin full, power interruption, or ice melt state)
- Flush level set point = Flush level setting 0-5.
- Flush used = 1-5. Flush level used in autoflush (0) mode.
- Water quality = An indication of water quality where 0-24 Extremely Mineral Laden; 25-30 Somewhat Mineral Laden; 31-65 Normal; 66-120 Very Clean; >120 Extremely Clean
- Error code - Diagnostic Error Code. Codes listed in software and on next page.
- Op mode = the current mode of the controller. Modes listed in software and on next page.
- Sump temperature = Reservoir water temperature in degrees F
- Discharge temperature = Discharge temperature in degrees F
- Supply voltage = approximation of AC voltage to the control board from the transformer.
- Bin stat = Bin thermostat, when used. 0 open, 1 closed
- RLO = SmartLock option. 0 not locked, 1 locked
- Ready to Harvest = Ice thickness sensor. 0 no ice, 1 ice
- Sump Full = Water level sensor. 0 no water, 1 water
- Sump Empty = Water level sensor. 0 no water, 1 water
- Remote = 0 not remote, 1 remote
- Curtain SW1 = 0 closed, 1 open
- Curtain SW2 = 0 closed, 1 open
- Water Solenoid = 0 off, 1 on
- Water Pump = 0 off, 1 on
- Hot Gas = 0 off, 1 on
- Condenser Fan/Aux = 0 off, 1 on
- Compressor = 0 off, 1 on
- Purge Valve = 0 off, 1 on
- Power up time= Time power connected to machine.
- Compressor run = Time compressor has been operating
- Power interrupts = Number of electrical power interruptions to the machine
- Bin setpoint = Set point of the Vari-Smart control
- Bin level (inches) = Ice level measured by the Vari-Smart control
- HGV counter = Number of times the hot gas valve has cycled. Equals harvest cycles.
- Auto flush level min = minimum level flush can be set to when set to automatic

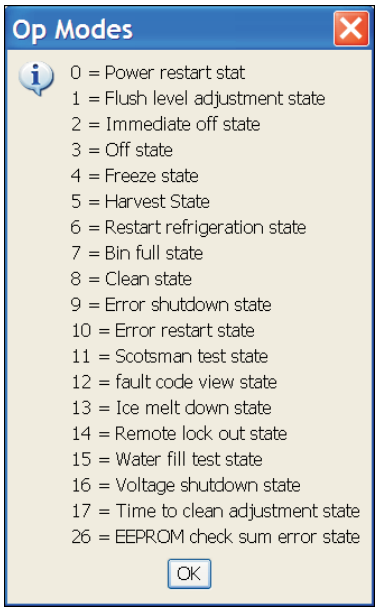
Reference



Example of generated PDF file



Error Code Display



Op Mode Display